

TAHIR SHUAIBU ABDULRAHMAN, Ph.D., M.ASME, R.COREN, MNSE,

Phone: +234 805 163 2954

Email: yatahir2301@gmail.com

Location: Nasarawa State, Nigeria

PROFESSIONAL SUMMARY: Advanced Manufacturing & Sustainable Engineering

Highly accomplished Mechanical Engineer and COREN-registered professional with a Ph.D. and over 15 years of combined experience in advanced manufacturing research, engineering education, and institutional leadership.

Critical & Philosophical Focus: Specialized in the intersection of Nanomaterials Science and Sustainable Manufacturing, driving national capacity development and import substitution. Expertise includes designing and optimizing Neem Oil-based nanofluids for demanding applications like machining oil and gas, automotive and aerospace-grade alloys (Ti6Al4V).

Practical Impact: Successful securing OF TETFUND research grants, leading curriculum review committee, and effectively translating complex research (machining optimization, thermal analysis) into publishable, industry-relevant knowledge. Dedicated to mentoring the next generation of Nigerian engineers.

RESEARCH EXPERTISE & TECHNICAL COMPETENCIES

Core Specialization

- Sustainable Machining: Optimization of machining processes (turning and drilling) for high-strength, difficult-to-machine alloys (Ti6Al4V) used in critical oil & gas and aerospace infrastructure.
- Nanomaterials Synthesis: Phytochemical and chemical synthesis, characterization, and integration of metal oxide nanoparticles (Nano CuO) into base fluids for enhanced thermal and tribological properties.
- Bio-Lubricants & Cutting Fluids: Development of indigenous, environmentally friendly bio-lubricants (Neem Oil-based) to replace imported petroleum-based fluids, promoting local content and reducing environmental hazards.

Software & Technical Skills

- Software: MATLAB (Advanced), Design Expert (Optimization), Origin, AutoCAD, CFTurbo (Hydraulics), Microsoft Office Suite.

- **Techniques:** Nanoparticle Synthesis & Advanced Characterization (XRD, FTIR, DLS, EDS), Thermal Analysis, Machining Process Optimization, Design and Analysis of Experiments (DOE).

PROFESSIONAL EXPERIENCE

Senior Lecturer, Federal Polytechnic Nasarawa, Department of Mechanical Engineering Technology

2024–Present (Post-Ph.D. Role)

- **Advanced Research Leadership:** Secured and managed TETFUND research project focusing on the Nano-CuO Neem Oil system for machining Titanium, directly contributing to national capacity in advanced lubrication technology.
- **Curriculum & Accreditation:** Chaired the Departmental Curriculum Review Committee, ensuring alignment of ND and HND programs with current industry demands and COREN accreditation standards.
- **Academic Excellence:** Designed and delivered advanced lectures in Strength of Materials, Fluid Mechanics, Thermodynamics, and Press & Cutting Tool Design to ND/HND students.
- **Mentorship:** Supervised over 20 final-year projects, successfully guiding students through the application of advanced concepts in manufacturing and design.

Lecturer, Department Secretary, Federal Polytechnic Nasarawa

2008–2024 (Roles leading to Senior Lecturer)

- Managed administrative functions as Department Secretary (2011–2015) and Project Coordinator (2024 to date), streamlining operational efficiency for over 300 students.
- Coordinated the Railway Engineering Technology unit (2014–2017), overseeing Curriculum Development and Offering Strategic Leadership.

INDUSTRIAL TRAINEE ENGINEER, Peugeot Automobile Nigeria

2005–2006 (Practical Engineering Exposure)

- Gained hands-on experience in component manufacturing and repair using conventional machines (lathe, milling, drilling).
- Provided maintenance support for diesel engines and materials handling equipment (forklifts).

RESEARCH & PUBLICATIONS

Key Research Projects

- Nano-CuO Neem Oil for Titanium Machining (TETFUND Funded, 2024–2025): Principal Investigator on a project developing a high-performance, indigenous bio-nanolubricant for machining aerospace-grade alloys (Ti6Al4V).
- Briquetting Plant Design (TETFUND, 2024): Contributed to sustainable energy research through the design and optimization of a biomass briquetting facility.
- Racing Car Cooling System Design (2025): Led a project focused on advanced thermal management and fluid dynamics.
- Comparative Analysis Of Temperature-Dependent Viscosity For Various Engine Oils To Determine Application Suitability (2025). Final year project supervised.

Selected Publications

- Neem Leaf-Mediated Green Synthesis of CuO Nanoparticles: A Sustainable Approach for Multidisciplinary Engineering Applications and Skills Development”. *7th Annual National Engineering Conference, 2025*.
- Synthesis, Thermal Characterization And Thermo-Physical Evaluation Of A Nano CuO-Infused Neem Oil Cutting Fluid For Enhanced Machining Of Ti-6Al-4valloy”: *Nigerian Journal Of Engineering, 2025*.
- “Phytochemical Synthesis, Characterisation and Size Optimisation Of Copper Nano Particles For Metal Cutting Coolant Applications”: *International Journal – Aaer Vol. 18 No.5 Jun-2022 ISSN: 2079-5074 International Journal Of Applied & Advanced Engineering Res. Vol. 18 No.5 Jun-2022 Issn: 2079-5074 MPRI-JAER, 2022*.
- “Lubricant Viscosity Analysis” — *NAS-LENS Journal, 2016*.

Academic Engagement

- Registered Engineer (COREN), Member, Nigerian Society of Engineers (MNSE), Member, American Society of Mechanical Engineers (ASME)
- Ph.D., Mechanical Engineering: Ahmadu Bello University, Zaria, Kaduna State, 2024 (Focus on Advanced Manufacturing/Tribology).

LEADERSHIP & SERVICE ROLES

- Secretary, Curriculum Review Committee (2014): successful introduction and addition of local contents to mechanical engineering program.
- Coordinator, Railway Engineering Technology (2014–2017): Managed the entire operational and academic delivery of a key discipline before accreditation of the program.

- Fiduciary & Institutional Stewardship (2016–2018): Demonstrated robust ethical and financial management while leading community endowment and fundraising committees, ensuring strict accountability of funds.
- Ethical Guidance & Mentorship (2021–Present): Served in high-visibility leadership roles focused on community education, ethical guidance, and formal mentorship.